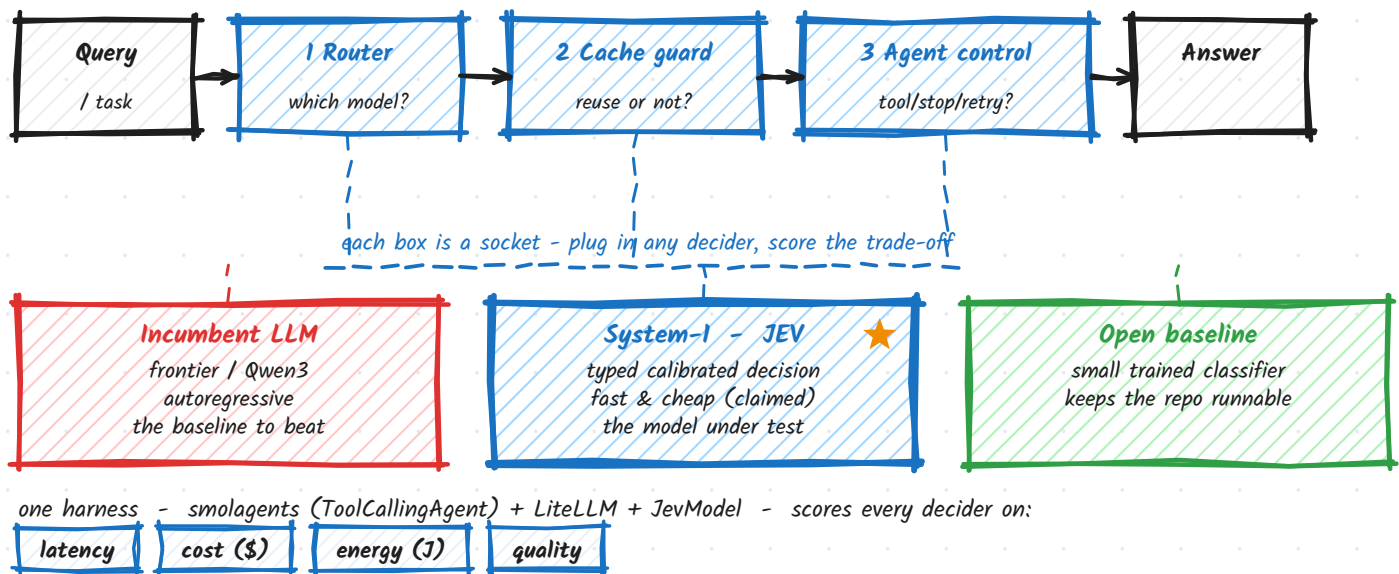


Deciding Without Generating

An efficiency benchmark of calibrated System-1 models vs. LLMs at the routing, caching & agent-control points.

arXiv preprint + repo New in ML @ NeurIPS MLSys 2027 6-8 wks -> early Nov

FIGURE 1 - decision points, interchangeable deciders & the shared harness



SUMMARY

Much of what LLMs do in real systems is not writing but DECIDING - which model answers, reuse a cached result?, which tool?, stop or retry? This project asks whether a calibrated System-1 model (Jev) can replace the LLM at those points, and measures the trade-off: latency, cost & energy saved vs. task quality lost, on one honest harness.

CONTRIBUTIONS

- 1) First independent, reproducible efficiency benchmark of the System-1 paradigm across three real LLM decision points - true wall-clock, dollar & energy cost, not token counts.
- 2) A drop-in JevModel adapter that makes a typed-output decision model interchangeable with chat LLMs behind one interface - reusable engineering.
- 3) An honest calibration + savings-vs-quality analysis - including where the vendor's efficiency claims do and don't hold (a contradiction there is the headline).

Deciding Without Generating ~ the build plan

THE FOUR DECIDERS

CEILING: Frontier LLM - GPT-5.x / Claude Opus / Gemini 3 Pro (via LiteLLM). Quality upper bound, used sparingly.

BASELINE: Qwen3-8B / 14B, self-hosted. The real cost competitor; keeps the repo runnable.

FLOOR: Small trained classifier. Open, fully reproducible.

UNDER TEST: Jev (System-1) via the JevModel adapter. Typed, calibrated decisions.

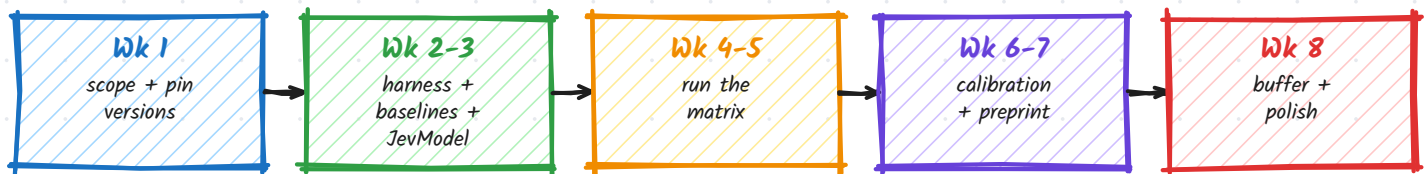
DECISION POINTS x TASKS

1 ROUTER: which model should answer? -> mixed-difficulty QA / routing set.

2 CACHE GUARD: reuse the cached output? -> QA / math over a semantic cache.

3 AGENT CONTROL: tool choice / stop / retry? -> tau2-bench (+ GAIA general check).

TIMELINE - a realistic 6-8 week build



PROTECT THE ARTIFACT - reproducibility & risk

Jev is proprietary, API-only, weights not released, and days old. Safeguards: (1) frame around the PARADIGM, Jev as instance one; (2) ship open baselines everywhere so the core result runs without the API; (3) treat vendor claims ("100x cheaper", "never hallucinates") as HYPOTHESES to test; (4) pin the model (jev-1.13.0) AND your gateway - LiteLLM had a March-2026 supply-chain incident, so isolate keys & use a known-good release.

WHERE IT GOES

arXiv preprint + clean GitHub repo FIRST & ALWAYS - it's what makes cold emails land. Then as they open: New in ML @ NeurIPS (poster), MLSys 2027, ICLR 2027 workshops. Verify all dates on each official site.

WHAT JEV IS

TypeSafe AI's first "System One" model (Sept 2026). Non-autoregressive; returns typed, calibrated decisions, not text. RLCD-trained. ~\$0.042/M in, output free; ~64k ctx; text-only; proprietary. Named after W.S. Jevons. VERIFY every spec - it's days old.